

Apurva Nakade

Assistant Teaching Professor, Department of Applied Mathematics & Statistics, Johns Hopkins University

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Experience	Assistant Teaching Professor, Johns Hopkins University	2026-Present
	Senior Lecturer, Johns Hopkins University	2023-2026
	Postdoctoral Lecturer, Northwestern University	2021-2023
	Instructor, Causeway Postbaccalaureate Program, Northwestern University	2022
	Postdoctoral Fellow, University of Western Ontario	2019-2021
	Academic Coordinator, Canada/USA Mathcamp	2018
	Mentor, Canada/USA Mathcamp	2017-2020
	Organizer and Mentor, Directed Reading Program, UWO and JHU	2017-2021
	Exotic Derivatives Trader, Nomura Capital India Pvt Ltd	2010
Education	Ph.D. in Mathematics, Johns Hopkins University	2019
	M.Sc. in Mathematics, Chennai Mathematical Institute	2013
	B.Tech. in Computer Science & Engineering, IIT Kanpur	2010
Teaching	Teaching Evaluations	
	Instructor, JHU AMS	
	SUMMER 2026	
	· Foundational Mathematics of Artificial Intelligence, Summer 2026	
	· Intermediate Probability and Statistics, Summer 2026	
	SPRING 2026	
	· Intro to Computational Math, Spring 2026	
	· Discrete Math, Spring 2026	
	· Monte Carlo Methods, Spring 2026	
	FALL 2025	
	· Monte Carlo Methods, Fall 2025	
	· Discrete Math, Fall 2025	
	SUMMER 2025	
	· Foundational Mathematics of Artificial Intelligence, Summer 2025	
	· Intermediate Probability and Statistics, Summer 2025	
	SPRING 2025	
	· Discrete Math, Spring 2025	
	· Monte Carlo Methods, Spring 2025	
	· Honors Algebra II, Spring 2025	
	· Intermediate Probability and Statistics, Spring 2025	

FALL 2024

- Discrete Math, Fall 2024
- Monte Carlo Methods, Fall 2024
- Honors Algebra II, Fall 2024

SUMMER 2024

- Engineering Innovation, Summer 2024

SPRING 2024

- Graph Theory, Spring 2024
- Linear Algebra and Differential Equations, Spring 2024
- Honors Algebra II, Spring 2024

FALL 2023

- Discrete Math, Fall 2023

Instructor, Northwestern

- Elementary Differential Equations, Spring 2023
- Foundations of Higher Mathematics, Winter 2023
- MENU Linear Algebra and Multivariable Calculus (Coordinator), 2022-23
- Introduction to Optimization, Winter, Spring 2022
- MENU Linear Algebra and Multivariable Calculus, 2021-22
- Single-Variable Differential Calculus, Fall 2021

Instructor, UWO

- Algebraic Topology (graduate level), Winter 2021
- Topology Bootcamp, Fall 2020
- Discrete Structures for Engineering, Fall 2020
- Calculus II for Mathematical and Physical Sciences, Winter 2020
- Calculus I for Mathematical and Physical Sciences, Fall 2019
- Topics in Category Theory, Fall 2019

Instructor, JHU

- Honors Single Variable Calculus, Fall 2018, 2017
- Symmetries & Polynomials, Intersession 2018
- Hitchhikers Guide to Algebraic Topology, Intersession 2017
- Differential Equations with Applications, Summer 2017, 2015
- Online Linear Algebra, Summer 2014

Instructor, Canada/USA Mathcamp

- How to Discover the Bell Curve, Summer 2026
- Theorem Proving in Lean, Summer 2020
- Galois Correspondence of Covering Spaces, Summer 2019
- From High School Arithmetic to Group Cohomology, Summer 2019
- Riemann Surfaces, Summer 2019
- Crash Course on Linear Algebra, Summer 2018
- How Curved Is a Potato?, Summer 2018
- Would I Ever Lie Group to You?, Summer 2018
- Cohomology via Sheaves, Summer 2018

Papers	String structures, 2-group bundles, and a categorification of the Freed-Quinn line bundle, Daniel Berwick-Evans, Emily Cliff, Laura Murray, Apurva Nakade, Emma Phillips, Quantum Symmetries, Contemporary Mathematics, vol. 813, Amer. Math. Soc., Providence, RI, pp. 257-301	2025
	Manifold Calculus and the h-principle, The Journal of Homotopy and Related Structures	2019
	Effect of increasing the energy gap between the two lowest energy states on the mixing time of the Metropolis algorithm (with Somenath Biswas), Information Processing Letters, IPL4801 (2012.08.012)	2012
Course Design & Materials	Introduction to Computational Mathematics, JHU	2026
	· A proof-based numerical analysis course on a free online textbook, with Jupyter notebook assignments and a group final project in place of a final exam	
	Visual Math Lab	2025-Present
	· A growing collection of interactive visualizations for teaching and learning mathematics	
	AMS Course Explorer, JHU	2025
	· An app for navigating the prerequisite graph of Applied Math & Statistics courses	
	Monte Carlo Methods, JHU	2025
	· An online textbook for a semester-long course on simulation, sampling and MCMC	
	Discrete Mathematics redesign, JHU	2023-Present
	· Homework resubmissions, frequent online assessments and restructured exams, built around an adapted open textbook	
	Introduction to Optimization, Northwestern	2022
	· Course notes in R Markdown and Excel modeling assignments for a course on linear programming and duality, rebuilt around applications	
	Open Textbook for Linear Algebra, Northwestern	2022
	· WeBWorK problem sets for a PreTeXt open educational resources textbook	
	Algebraic Topology, UWO	2021
	· Problem sets for a graduate course on the fundamental group, covering spaces and singular homology	
	Formalizing mathematics in Lean	2020
	· Contributions to mathlib's surreal numbers and convex optimization libraries	
	Discrete Structures for Engineering, UWO	2020
	· Rebuilt for fully asynchronous delivery: weekly video lectures, hundreds of WeBWorK problems written as a team, and a Piazza forum; presented at a conference on e-assessment in the mathematical sciences	
	Honors Single Variable Calculus, JHU	2017-2018
	· Two semesters of calculus in one, taught inquiry-based in a flipped classroom	
	Intersession courses for non-math majors, JHU	2017-2018
	· Symmetries & Polynomials: from group theory to the unsolvability of the quintic	
	· Hitchhiker's Guide to Algebraic Topology: algebraic topology and its applications	

	Canada/USA Mathcamp courses	2017-2026
	· Week-long courses for advanced high school students, including Theorem Proving in Lean, How to Discover the Bell Curve, Riemann Surfaces, Covering Spaces and Cohomology via Sheaves	
Grants & Awards	Open Educational Resources Faculty Grant (joint with Dhruv Azad), JHU	2026
	· \$5000 to develop interactive visualization apps for computational math courses	
	Open Educational Resources Faculty Grant (joint with Aaron Greicius), NU	2022
	William Kelso Morrill Award for Excellence in Mathematics, JHU	2019
	Finalist for the KSAS Excellence in Teaching Awards, JHU	2019
	Prof. Joel Dean Award for Excellence in Teaching in Mathematics, JHU	2016
	AMS Graduate Student Travel Grant	2019
Service	Member of DS Masters admissions committee in JHU AMS department	2025-2026
	Helped connect undergraduate students with research mentors in JHU AMS department	2024-2025
	Judge for poster presentations at JHU Matrix Conference	2025
	Judge for poster presentations at MAA MD-DC-VA Section Meeting	Fall 2024
	Supplementary Instructor for the Causeway Postbaccalaureate Program, NU	2022
	Directed Reading Program, UWO, JHU	2017-2021
	· Started DRP at UWO in Fall 2019 · Organizer and mentor for DRP at JHU and UWO Professional Development	
Professional Development	Faculty Forward Fellowship, JHU	2025
	Introduction to Education Research Workshop, JHU	2023
	MAA Section NExT Fellow, 2023 cohort	2023-25
	Mathematical Association of America Member	2020-Present
	MAA Project NExT Fellow, Brown'20 cohort	2020
	Teaching Academy Certification, JHU	2019
	Participated in several workshops by the Center of Teaching & Learning at UWO	2019-2020
	Participated in the MSRI Critical Issues in Mathematics Education Workshop	2022
	Participated in the MAA Modeling Inspiration for Differential Equations Workshop	2022
	Participated in online Mastery Grading Workshop	2019
	Attended the Science of Learning Symposium, JHU	2014-2018

Talks	Discrete Chern–Simons via 2-group bundles on elliptic curves, CMS Session on Homotopy Theory	2020
	What is a Spectrum?, University of Western Ontario, Basic Notions Seminar	2020
	Manifold calculus and the h -principle	2017-2019
	· University of Regina, Geometry & Topology Seminar, 2019	
	· University of Western Ontario, Geometry & Topology Seminar, 2019	
	· University of Rochester, Topology Seminar, 2019	
	· Workshop on Functor Calculus, Ohio State University, 2019	
	· Spaces of Embeddings, BIRS, Banff, 2019	
	· AMS Special Session in Homotopy Theory, UC Riverside, 2017	
	Weiss fibration sequence, MIT Talbot Workshop	2019
	Constructing a Homotopy Type for Triply-Graded Link Homology, AMS Sectional Meeting, University of Hawaii	2019
Conferences Attended	Homotopy colimits and limits, European Autumn School in Topology	2017
	MAA MD-DC-VA Section Meeting	Spring 2024
	MAA MD-DC-VA Section Meeting	Fall 2023
	United States Conference on Teaching Statistics	2023
	MSRI CIME Workshop	2022
	Project NExT at MAA Mathfest, Online	2021
	Lean for the curious mathematician	2020
	Spaces of Embeddings, BIRS, Banff	2019
	MSRI Summer School, Cortona, Italy	2019
	MRC Workshop, Providence RI	2019
	MIT Talbot Workshop, Austin TX	2019
	Workshop on Functor Calculus, Ohio State University	2019
	AMS Sectional Meeting, University of Hawaii	2019
	Arizona Winter School, Arizona State University	2019
	Joint Mathematical Meetings, Baltimore	2019
	Symplectic Geometry and Homotopy Theory, UCLA	2018
	MSRI Summer School, Fields Institute, Toronto	2018
	Graduate Student Conference, Temple University	2018
	AMS Sectional Meeting, UC Riverside	2017
	European Autumn School in Topology, Netherlands	2017
	Topology Festival, Cornell University	2017
	Georgia International Topology Conference, University of Georgia	2017
	Alpine Algebraic & Applied Topology Conference, Switzerland	2016
	WCATSS, University of Oregon, Eugene	2016
	GSTSC, Indiana University, Bloomington	2016
	Mid-Atlantic Topology Conference, Johns Hopkins University	2016
	Midwest Topology Seminar, Northwestern University	2016
	Geometry and Topology Conference, Lehigh University	2015
	Mid-Atlantic Topology Conference, University of Virginia	2015
	Midwest Topology Seminar, University of Illinois Chicago	2015

Midwest Topology Seminar, Northwestern University	2015
Modular invariants in Topology and Analysis, Regensburg	2015
WCATSS on Field theories, UBC	2015
Introductory Workshop on Algebraic Topology, MSRI	2015
Joint Mathematical Meetings, Baltimore	2015
Classification of Manifolds, NEHU	2013
H-principle, Chennai Mathematical Institute	2012
Groups and geometries, ISI, Bangalore	2012
String Topology, Vivekananda University	2012
Kervaire Invariant One, ISI, Kolkata	2012
Number Theory workshop, Tezpur University	2011
Lie algebras and their representations, CMI	2011
Nurture camp, Institute of Mathematical Sciences	2007
IMO, Slovenia	2006
IMOTC, HBCSC	2003-2006
Kishore Vaigyanik Protsahan Yojana (KVPPY)	2005